

Summary report

Wellbore: 15/9-F-4

Period: 2008-04-04 00:00 - 2008-04-05 00:00

Status:	normal
Report creation time:	2018-05-03 13:52
Report number:	89
Days Ahead/Behind (+/-):	
Operator:	StatoilHydro
Rig Name:	MÆRSK INSPIRER
Drilling contractor:	Mærsk Contractors
Spud Date:	2007-03-10 00:00
Wellbore type:	
Elevation RKB-MSL (m):	54.9
Water depth MSL (m):	91
Tight well:	Y
HPHT:	Y
Temperature (I):	
Pressure (I):	
Date Well Complete:	2008-02-28

Dist Drilled (m):	-999.99
Penetration rate (m/h):	-999.99
Hole Dia (I):	
Pressure Test Type:	formation integrity test
Formation strength (g/cm3):	1.65
Dia Last Casing (I):	

Depth at Kick Off mMD:	
Depth at Kick Off mTVD:	
Depth mMd:	3510
Depth mTVD:	3138.8
Plug Back Depth mMD:	3448
Depth at formation strength mMD:	2760
Depth At Formation Strength mTVD:	2599
Depth At Last Casing mMD:	3508
Depth At Last Casing mTVD:	3136.9

Summary of activities (24 Hours)

Changed die holders in TDS. Ran adjustment sub activation tool and retracted adjustment sub. Tested metal seal in adjustment sub to 380 bar. Rigged up casing running equipment to run hanger RT. Observed dropped object on drillfloor - suspended operations for investigation and repair work on aft PRS. RIH with hanger RT on 10 3/4" casing.

Summary of planned activities (24 Hours)

Set 10 3/4 MS seal.

Operations

Start time	End time	End Depth mMD	Main - Sub Activity	State	Remark
00:00	00:15	119	drilling -- casing	ok	Picked up pre-made 5 1/2" HWDP (LS-1) with 10 3/4" Itco spear with PRS. Attempted to make up stand to TDS - negative - slipped in TDS back up dies.
00:15	01:15	119	interruption -- other	ok	Broke off TDS from stand. Set down stand in PRS. Lowered TDS and inspected torque wrench dies, removed swarf from dies. Re-attempted make up to stand in TDS - ok.
01:15	01:30	139	drilling -- casing	ok	Ran 10 3/4" Itco spear 3 m into 10 3/4" tie-back casing and latched same. Took up/down weight 82/79 MT. RIH with tie-back casing on 5 1/2" HWDP (LS-1). Put meter marks from casing hanger landing shoulder. Tagged with 4 MT at ~139,4 m, i.e. 30-40 cm shallow - no colour marks observed in WH B-annulus outlet. Re-attempted tag - same depth.
01:30	02:30	139	drilling -- casing	ok	Evaluated situation. Checked drawings for potential hang-ups. Checked tally and joints left on deck - ok. Made two attempt to index pipe before tagging again - same depth. Sudden overpull of 14 MT observed after pulling some 30 cm above tag depth.
02:30	03:15	139	drilling -- casing	ok	Decided to attempt to screw in 1-2 turn in order to verify tie-back adapter depth. RIH and tagged approx 10 cm deeper than earlier. Observed lower orange and white band in B-annulus outlet. Measured required distance for adjustment sub travel (15") and decided to continue according to plan.
03:15	04:15	139	drilling -- casing	ok	Set torque limit to 2700 Nm. Made 3 attempts to enter tie-back adapter. Entered tie-back adapter on 4th attempt and observed weight increasing about 3-4 MT for each turn. Slacked off to neutral (80 MT) for each turn achieved. String stalled out at 2700 Nm after 5+ turns achieved. Took 10 MT overpull. Rotated at 1 rpm. Attempted make up to 10 000 Nm (optimum torque) - spear started to slip at 5500 Nm. Re-engaged spear. Adjusted torque setting to minimum torque 6600 Nm -spear started to slip at 5500 Nm. Re-engaged spear. Set torque limit to 10 000 Nm, increased rotation to 3 RPM. String stalled out on 10 000 NM (optimum torque) - ok.
04:15	04:45	139	drilling -- casing	ok	Took 30 MT overpull to check pre-tension in casing towards datum. Stretched pipe approx 2" and observed lower white band through B-annulus outlet.
04:45	06:00	139	drilling -- casing	ok	Rigged down Odfjell casing tong and removed PS-30 slips. Installed masterbushing and bowl.
06:00	09:00	0	interruption -- other	ok	Jammed torque wrench on TDS. Inspected equipment and found that wrong die-holders had been installed. Changed die-holders.
09:00	09:15	0	drilling -- casing	ok	Held toolbox talk prior to picking up adjustment sub activation tool.
09:15	10:15	28	drilling -- casing	ok	RIH with adjustment sub and located same at 28,4 m MD. Checked serial numbers on pipe. Made up TDS.
10:15	11:00	28	drilling -- casing	ok	Pressured up on drillpipe to 140 bar to activate horizontal dogs. Pulled back 0,4 m and confirmed engaged in vertical dog profile by 9 MT overpull. Rotated string with 22 righthand turns. Observed colored band through B-annulus outlet to confirm adjustment sub collapsing. Increased overpull to 30 MT and continued rotating until TDS stalled out on 9 kNm and overpull reduced. Slacked off string to 28,4 m MD mark and waited for 2 min. Bleed off pressure on DP. Obtained a total of 17" travel on adjustment sub.
11:00	11:45	0	drilling -- casing	ok	POOH with adjustment sub activation tool and racked back in derrick.
11:45	12:00	0	drilling -- casing	ok	Made up Drillquip straddle cup tester.
12:00	12:30	28	drilling -- casing	ok	RIH with straddle cup tester and located same at 28,4 m MD across metal seal in adjustment sub. Meanwhile held toolbox talk for pressure testing.
12:30	13:00	28	drilling -- casing	ok	Lined up cement unit on standpipe. Performed line test to 380 bar.
13:00	14:30	28	drilling -- casing	ok	Pressure tested adjustment sub (metal seal) to 380 bar / 10 min - ok. Kept B-annulus open throughout test.
14:30	15:00	140.5	drilling -- casing	ok	RIH with straddle cup tester from 28,4 m and located same across tie-back adapter at 140,5 m MD.
15:00	16:00	140.5	drilling -- casing	ok	Filled pipe with approx 1,8 m3 packer fluid prior to testing. Pressured up to 380 bar - observed leakage from B-annulus outlet - not able to obtain test.
16:00	16:15	130.5	drilling -- casing	ok	Pulled back 10 m to put cup tester in blank pipe to test tool. Pressured up - not able to obtain test - straddle cup test leaking.
16:15	17:30	0	drilling -- casing	ok	POOH with straddle cup tester from 130,5 m MD to surface. Laid out cup tester and inspected same. Found no tears or flaws, but OD reduced significantly (9,85"/9,70" vs originally 10,13")
17:30	17:45	0	drilling -- casing	ok	Held toolbox talk prior to rigging casing running equipment.
17:45	18:00	0	drilling -- casing	ok	Rigged up casing running equipment. Found a yellow sheared allen bolt on walkway along drillfloor. Suspended operation and evaluated situation.
18:00	20:00	0	interruption -- other	ok	Isolated aft PRS and TDS and inspected same with manrider. Found origin of bolthead to be from backside of aft PRS upper grip head.
20:00	00:00	0	interruption -- other	ok	Parked aft PRS for repair. Performed repair work on aft PRS. Broke out and re-tightened in total 24 allen/hex bolts with prescribed momentum (600 ft lbs). Replaced broken bolt. Meanwhile : -Installed BX3 elevator in bails -Brought Vetco/BOT equipment to cantilever

Equipment Failure Information

Start time	Depth mMD	Depth mTVD	Sub Equip - Syst Class	Operation Downtime (min)	Equipment Repaired	Remark
00:00	0		miscellaneous equ syst -- other	0	00:00	Troubleshoot programming on casing tong.
00:00	119		hoisting equ -- top drive	0	00:00	Unable to make up TDS with pipe iin PRS - pipe slipped in torque wrench
00:00	0		pipe handling equ syst -- vertical pipe handling equ syst	0	00:00	Broke bolt head on PRS grip head. Falling object.

Pore Pressure

Time	Depth mMD	Depth TVD	Equ Mud Weight (g/cm3)	Reading
00:00	3510		1.08	measured